

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Design a machine learning solution for predicting customer loan approval using a suitable learning algorithm.

Answer

Customer loan approval is a **binary classification problem** because the output is either **Approved (1) or Rejected (0)**. A suitable algorithm is **Logistic Regression**.

Stages of Model Development

1. Problem Definition

Predict whether a customer's loan should be approved.

2. Data Collection

Collect customer details such as:

- Age
- Income
- Credit Score
- Employment Status
- Loan Amount
- Previous Loan History

3. Data Preparation

- Remove missing values.
- Remove duplicate records.

- Encode categorical variables.
- Normalize numerical values.
- Split dataset into 80% training and 20% testing.

4. Model Training

Train the Logistic Regression model using the training dataset to learn patterns.

5. Prediction

The trained model predicts whether a new customer's loan will be approved.

6. Performance Evaluation

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Flowchart

Customer Data



Data Preprocessing



Feature Selection



Train-Test Split



Model Training

(Logistic Regression)



Prediction



Performance Evaluation

Advantages

- Fast decision making.
- Reduces manual errors.
- Improves loan approval accuracy.

Conclusion

Machine learning automates loan approval by learning from historical customer data and providing accurate predictions.

Q2. Describe the architecture of an Artificial Neural Network (ANN).

Answer

An **Artificial Neural Network (ANN)** is a computational model inspired by the human brain. It consists of interconnected neurons arranged in layers.

Architecture

1. Input Layer

Receives input features.

Example:

- Age
- Income
- Credit Score

2. Hidden Layer

Processes inputs using:

- Weights
- Bias
- Activation Function

3. Output Layer

Produces the final prediction.

Example:

- Loan Approved
- Loan Rejected

Components

Artificial Neuron

Computes

$$z = \sum w_i x_i + b$$

where

- x = input
- w = weight
- b = bias

Output

$$y = f(z)$$

Weights

Determine the importance of each input.

Bias

Shifts the activation function to improve learning.

Activation Function

Introduces non-linearity.

Examples:

- Sigmoid
- ReLU
- Tanh

Diagram

Inputs

x1 x2 x3

\ | /

\ | /

[Hidden Layer]

|

[Output Layer]

Working

1. Input enters ANN.
2. Hidden layer performs weighted calculations.
3. Activation function generates output.
4. Output layer gives prediction.

Advantages

- Learns complex relationships.
- High prediction accuracy.
- Suitable for classification and regression.

Conclusion

ANN mimics the human brain by processing information through interconnected neurons to make intelligent predictions.

Q3. Analyze the working of the Perceptron learning algorithm.

Answer

The **Perceptron** is the simplest neural network used for binary classification.

Steps

Step 1

Initialize weights and bias.

Example

$w_1 = 0.2$

$w_2 = 0.3$

$b = 0.1$

Learning Rate = 0.1

Input

$$x_1 = 1$$

$$x_2 = 0$$

$$\text{Target} = 1$$

Step 2

Calculate net input.

$$\begin{aligned} z &= (w_1x_1 + w_2x_2) + b \\ &= (0.2)(1) + (0.3)(0) + 0.1 \\ &= 0.3 \end{aligned}$$

Step 3

Activation Function

If

$$z > 0$$

$$\text{Output} = 1$$

$$\text{Otherwise Output} = 0$$

$$\text{Predicted Output} = 1$$

Error

$$\begin{aligned} \text{Error} &= \text{Target} - \text{Output} \\ &= 1 - 1 = 0 \end{aligned}$$

No weight update is required.

Weight Update Formula

$$w = w + \eta(\text{error})x$$

Bias

$$b = b + \eta(\text{error})$$

Flowchart

Input



Weighted Sum



Activation Function



Prediction



Error Calculation

↓

Weight Update

Advantages

- Simple.
- Fast.
- Easy implementation.

Limitation

Works only for linearly separable data.

Conclusion

The perceptron learns by adjusting weights until classification errors become minimum.

Q4. Illustrate Forward Propagation in an MLP.

Answer

Forward propagation transfers information from the input layer to the output layer.

Given

Inputs

$$x_1 = 1$$

$$x_2 = 2$$

Weights

$$w_1 = 0.5$$

$$w_2 = 0.4$$

Bias

$$b = 0.2$$

Activation Function

Sigmoid

Step 1

Weighted Sum

$$\begin{aligned} z &= (1)(0.5) + (2)(0.4) + 0.2 \\ &= 0.5 + 0.8 + 0.2 \\ &= 1.5 \end{aligned}$$

Step 2

Sigmoid Function

$$\begin{aligned} \sigma(z) &= \frac{1}{1 + e^{-z}} \\ &= \frac{1}{1 + e^{-1.5}} \\ &= 0.817 \end{aligned}$$

Output

0.817

Forward Propagation Flow

Inputs

↓

Weighted Sum

↓

Bias Added

↓

Activation Function

↓

Output

Advantages

- Produces predictions.
- Used in every neural network.
- Supports deep learning models.

Conclusion

Forward propagation computes outputs by passing inputs through weighted connections and activation functions, enabling prediction in multilayer neural networks.

Q1. A coin is tossed 10 times. Estimate the probability of Heads using MLE.

Given

- Total number of tosses, **n = 10**
- Number of Heads, **H = 7**

Formula

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is:

$$\boxed{\hat{p} = 0.7}$$

Q2. Out of 50 students, 42 passed. Find the MLE of the probability of passing.

Given

- Total students = 50
- Students passed = 42

Formula

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Observations}}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer

The Maximum Likelihood Estimate (MLE) of passing is:

$$\boxed{\hat{p} = 0.84}$$

Q3. A factory produces 100 bulbs, and 8 are defective. Find the MLE of defect probability.

Given

- Total bulbs = 100
- Defective bulbs = 8

Formula

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer

The Maximum Likelihood Estimate (MLE) of a defective bulb is:

$$\boxed{\hat{p} = 0.08}$$

Q4. A biased die is rolled 60 times and 6 appears 18 times. Estimate the probability using MLE.

Given

- Total rolls = 60
- Number of sixes = 18

Formula

$$\hat{p} = \frac{\text{Number of Times 6 Appears}}{\text{Total Rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.30$$

Answer

The Maximum Likelihood Estimate (MLE) of rolling a 6 is:

$$\boxed{\hat{p} = 0.30}$$

Q5. Gradient Descent Weight Update**Given**

- Input, $x = 4$
- Weight, $w = 2$
- Actual Output = 10
- Gradient = 4
- Learning Rate, $\eta = 0.01$

(i) Predicted Output**Formula**

$$y = w \times x$$

Calculation

$$y = 2 \times 4 = 8$$

Answer

$$\boxed{\text{Predicted Output} = 8}$$

(ii) Error**Formula**

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation

$$\text{Error} = 10 - 8 = 2$$

Answer

$$\boxed{\text{Error} = 2}$$

(iii) Updated Weight

Formula

$$\text{New Weight} = \text{Old Weight} - (\eta \times \text{Gradient})$$

Calculation

$$= 2 - (0.01 \times 4)$$

$$= 2 - 0.04$$

$$= 1.96$$

Answer

Updated Weight = 1.96

Q6. A neuron has two inputs $x_1 = 2$, $x_2 = 3$, weights $w_1 = 0.5$, $w_2 = 0.4$, and bias $b = 1$.

Find the output using a linear activation function.

Given

- $x_1 = 2$, $x_2 = 3$
- $w_1 = 0.5$, $w_2 = 0.4$
- Bias $b = 1$

Formula

$$y = (x_1 w_1 + x_2 w_2) + b$$

Calculation

$$y = (2 \times 0.5) + (3 \times 0.4) + 1$$

$$= 1 + 1.2 + 1$$

$$= 3.2$$

Answer

Neuron Output = 3.2

Q7. A perceptron has inputs $x_1 = 1$, $x_2 = 0$, weights $w_1 = 0.6$, $w_2 = 0.2$, bias $b = -0.3$.

Find the output using the step activation function.

Given

- $x_1 = 1$, $x_2 = 0$
- $w_1 = 0.6$, $w_2 = 0.2$
- Bias $b = -0.3$

Step 1: Net Input

Formula

$$z = x_1w_1 + x_2w_2 + b$$

Calculation

$$\begin{aligned} z &= (1 \times 0.6) + (0 \times 0.2) - 0.3 \\ &= 0.6 - 0.3 \\ &= 0.3 \end{aligned}$$

Step 2: Apply Step Activation Function

Rule:

- If $z > 0$, Output = 1
- Otherwise, Output = 0

Since,

$$0.3 > 0$$

Answer

Perceptron Output = 1

Q8. A Multilayer Perceptron has input $x = 2$, weight $w = 1.5$, and bias $b = 0.5$. Calculate the output after forward propagation using a linear activation function.

Given

- Input $x = 2$
- Weight $w = 1.5$
- Bias $b = 0.5$

Formula

$$y = (w \times x) + b$$

Calculation

$$\begin{aligned} y &= (1.5 \times 2) + 0.5 \\ &= 3 + 0.5 \\ &= 3.5 \end{aligned}$$

Answer

Forward Propagation Output = 3.5

Q9. A neural network has Actual Output = 1, Predicted Output = 0.8, Learning Rate = 0.05, Gradient = 0.6, Current Weight = 2.5. Find the error and updated weight.

Given

- Actual Output = 1
- Predicted Output = 0.8
- Learning Rate = 0.05
- Gradient = 0.6
- Current Weight = 2.5

(i) Error

Formula

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation

$$= 1 - 0.8 = 0.2$$

Answer

$$\boxed{\text{Error} = 0.2}$$

(ii) Updated Weight

Formula

$$\text{New Weight} = \text{Old Weight} - (\eta \times \text{Gradient})$$

Calculation

$$\begin{aligned} &= 2.5 - (0.05 \times 0.6) \\ &= 2.5 - 0.03 \\ &= 2.47 \end{aligned}$$

Answer

$$\boxed{\text{Updated Weight} = 2.47}$$

Q10. Compare Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent.

Batch Gradient Descent	Stochastic Gradient Descent (SGD)	Mini-Batch Gradient Descent
Uses the entire dataset for one update.	Uses one training sample for each update.	Uses a small batch of samples for each update.
Slow but stable convergence.	Fast updates but noisy convergence.	Faster and more stable than SGD.
High memory requirement.	Low memory requirement.	Moderate memory requirement.
Suitable for small datasets.	Suitable for very large datasets.	Most commonly used in deep learning.

Code of Conduct:

I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation